

Business Intelligence Project



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Overview

This Business Intelligence (BI) project centers around utilizing sample data from "Foodmart" CSV files to develop a data analysis framework. The project involves importing data into Microsoft SQL Server 2022 Express and the Excel Data Model via Power Query. The key objectives include cleaning, transforming, and consolidating data, establishing relational integrity across datasets, and leveraging tools like SQL Server, Excel, and Power BI to enable advanced analytics.

The dataset comprises eight tables, including customer, product, region, store, and calendar lookup tables, alongside transaction and returns data for 1997–1998. Enhancements like calculated columns and relationships have been carefully implemented to optimize analysis and reporting capabilities.

The project has incorporated advanced data modeling features, interactive visualizations, and DAX measures to perform time-based analysis (e.g., Year-to-Date, Quarter-to-Date). These features empower the business to derive actionable insights, track trends, and drive strategic decision-making.

Objectives

- **Data Consolidation:** Integrate and centralize data from various sources (CSV files) into a unified platform, ensuring consistency and referential integrity.
- **Enhanced Analytical Capabilities:** Utilize calculated columns, DAX measures, and improved lookup tables (e.g., calendar with fiscal periods) to support detailed and time-based analysis.
- **Interactive Dashboards:** Create user-friendly dashboards in Power BI and Excel that enable stakeholders to visualize key metrics like net revenue, transaction trends, and regional sales performance.
- **Operational Insights:** Streamline analysis of transaction and returns data to optimize inventory management and reduce revenue leakage.
- **Time Intelligence:** Employ time-based metrics (MTD, QTD, YTD) to better understand seasonal and temporal patterns in sales performance.
- **Stakeholder Empowerment:** Design intuitive filters and slicers for end-users to explore data dynamically, facilitating data-driven decision-making.
- **Integration Across Tools:** Ensure seamless data flow and compatibility between SQL Server, Excel, and Power BI for a holistic and flexible analysis environment.
- **Performance Optimization:** Monitor and improve operational efficiency by analyzing sales trends, return patterns, and customer behaviors.

Dataset Details

The dataset consisted of eight tables in CSV format:

- **Lookup Tables:** Customers, products, regions, stores, and general calendar.
- **Data Tables:** Transaction data for 1997, transaction data for 1998, and returns data covering 1997–1998.

All tables were successfully imported into SQL Server 2022 Express and Excel's data model. Relationships were carefully created between the various tables to ensure proper data flow and to maintain referential integrity.

Active	Table 1	Cardinality	Filter Direction	Table 2
Yes	Returns [product_id]	Many to One (*:1)	<< To Returns	Product_Lookup [product_id]
Yes	Returns [return_date]	Many to One (*:1)	<< To Returns	Calendar_Lookup [date]
Yes	Returns [store_id]	Many to One (*:1)	<< To Returns	Store_Lookup [store_id]
Yes	Store_Lookup [region_id]	Many to One (*:1)	<< To Store_Lookup	Region_Lookup [region_id]
Yes	transactions [customer_id]	Many to One (*:1)	<< To transactions	Customer_Lookup [customer_id]
Yes	transactions [product_id]	Many to One (*:1)	<< To transactions	Product_Lookup [product_id]
No	transactions [stock_date]	Many to One (*:1)	<< To transactions	Calendar_Lookup [date]
Yes	transactions [store_id]	Many to One (*:1)	<< To transactions	Store_Lookup [store_id]
Yes	transactions [transaction_date]	Many to One (*:1)	<< To transactions	Calendar_Lookup [date]

Data Transformation and Processing

- **Leveraging Power Query**, all tables were cleaned and standardized for formatting and appropriate data types.
- **Custom-calculated columns** were created to enhance the dataset and enable additional pivot table modeling for deeper analytical insights.
- The transaction tables for 1997 and 1998 were merged into a single consolidated table within Power Query to streamline analysis and reporting.

Example: Calendar Lookup Table, original state:

Upon import, the calendar lookup table contained only basic short date information. This limitation restricted its usability for advanced analysis and reporting tasks.

	A	B	C	D
1	date			
2		1/1/1997		
3		1/2/1997		
4		1/3/1997		
5		1/4/1997		
6		1/5/1997		
7		1/6/1997		
8		1/7/1997		
9		1/8/1997		
10		1/9/1997		

Calendar Lookup Table, current state:

Significant enhancements were made to the table to provide greater analytical flexibility for use in SQL queries, pivot charts, and Power BI reports or dashboards. These improvements include the addition of calculated columns, such as:

- **Day, Month, and Year Columns:** Extracted individual components of the date for more detailed breakdowns.
- **Quarter Information:** Identified fiscal or calendar quarters, enabling trend and seasonal analysis.
- **End-of-Month and Year Dates:** Provided clear period-end markers to streamline calculations and time-based queries.
- **Weekend/Weekday Indicator:** Differentiated workdays from weekends, enhancing insights into activity patterns and performance trends.

These changes transformed the calendar lookup table into a robust and versatile resource, empowering comprehensive and dynamic data analysis across various tools and platforms.

The screenshot displays an Excel Data Model table view. A callout box highlights the first two rows of the table, showing columns: date, Year, Month, Month Name, Quarter, Start of Week, Day, Day Name, weekend, and qtr. The data for these rows is as follows:

date	Year	Month	Month Name	Quarter	Start of Week	Day	Day Name	weekend	qtr
7/1/1997 12:00:00 AM	1997	7	July	3	6/30/1997 12:00:00 AM	1	Tuesday	N	Q3
7/2/1997 12:00:00 AM	1997	7	July	3	6/30/1997 12:00:00 AM	2	Wednesday	N	Q3

The main table view below shows a detailed list of data rows with columns including date, Year, Month, Month Name, Quarter, Start of Week, Day, Day Name, weekend, qtr, name, end_of_month, date (Year), date (Quarter), date (Month Index), date (Month), and Add Columns. The table contains 31 rows of data for July 1997, followed by summary rows for QTD total trans, QTD Trans, MTD Trans, last month trans, and Month over Month Trans.

Excel Data Model

Visualizing Table Relationships and Structure

The screenshot offers a comprehensive view of the relationships between the various tables, detailing cardinality and filter direction to illustrate how data flows and interacts within the model. This visual representation highlights:

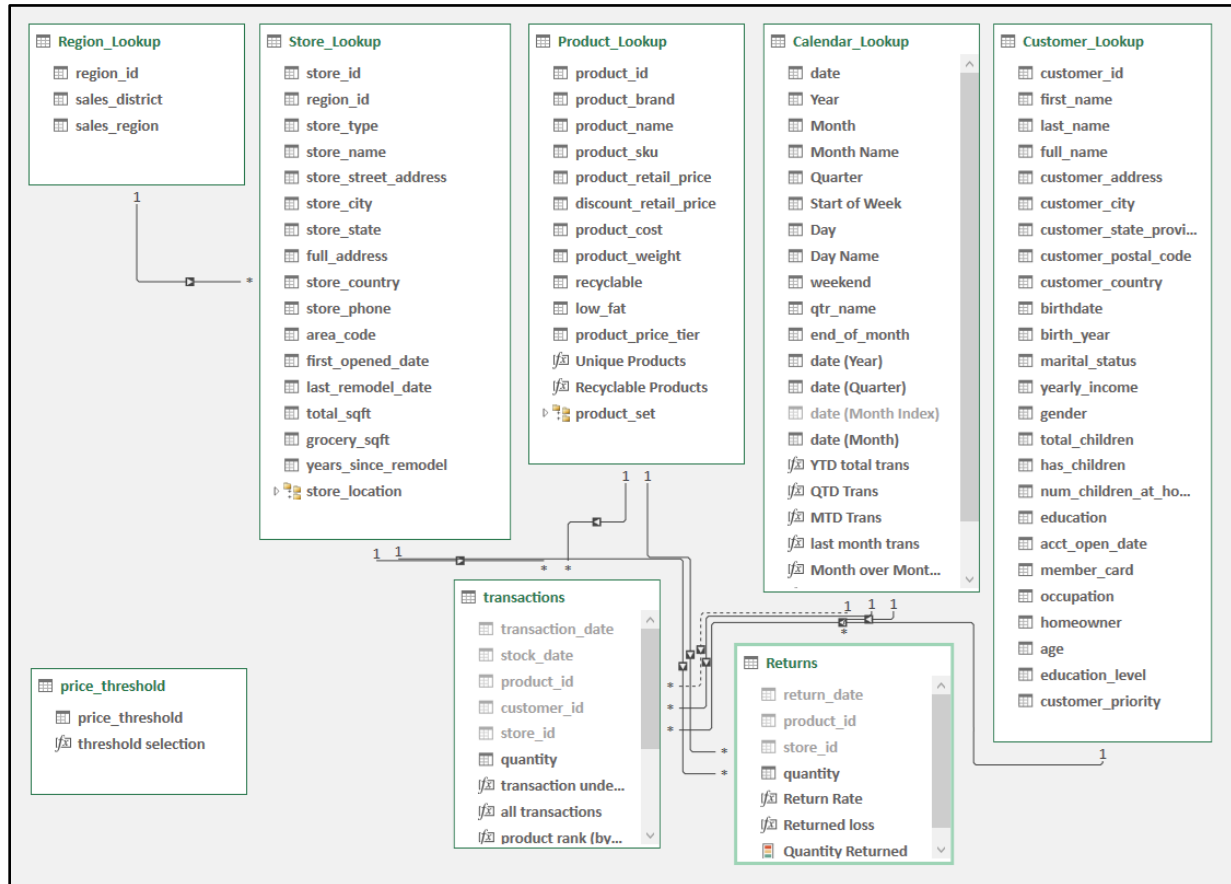
- **Table Relationships:** Clearly defined connections between lookup tables and data tables, ensuring proper relational integrity across the dataset.
- **Cardinality:** Specifies the nature of relationships, such as one-to-one or one-to-many, providing insight into the structure of the data model.
- **Filter Direction:** Demonstrates the pathways along which data propagates, aiding in understanding dependencies and interactions between tables.

In addition, the table view includes:

- **Calculated Columns:** Custom columns designed to extend analytical capabilities and enhance reporting potential.
- **Custom Measures:** Tailored metrics applied to meet specific business intelligence objectives and uncover actionable insights.
- **Hierarchies:** Structured groupings within tables to facilitate intuitive drill-down analysis in pivot charts and dashboards.

- **Hidden Foreign Keys:** Greyed-out fields representing foreign keys that are concealed to streamline the model's visual presentation while maintaining functionality.

This level of detail ensures the dataset is well-optimized for advanced analysis across tools like SQL Server, Excel, and Power BI.



Excel Pivot Table and Associated Visuals

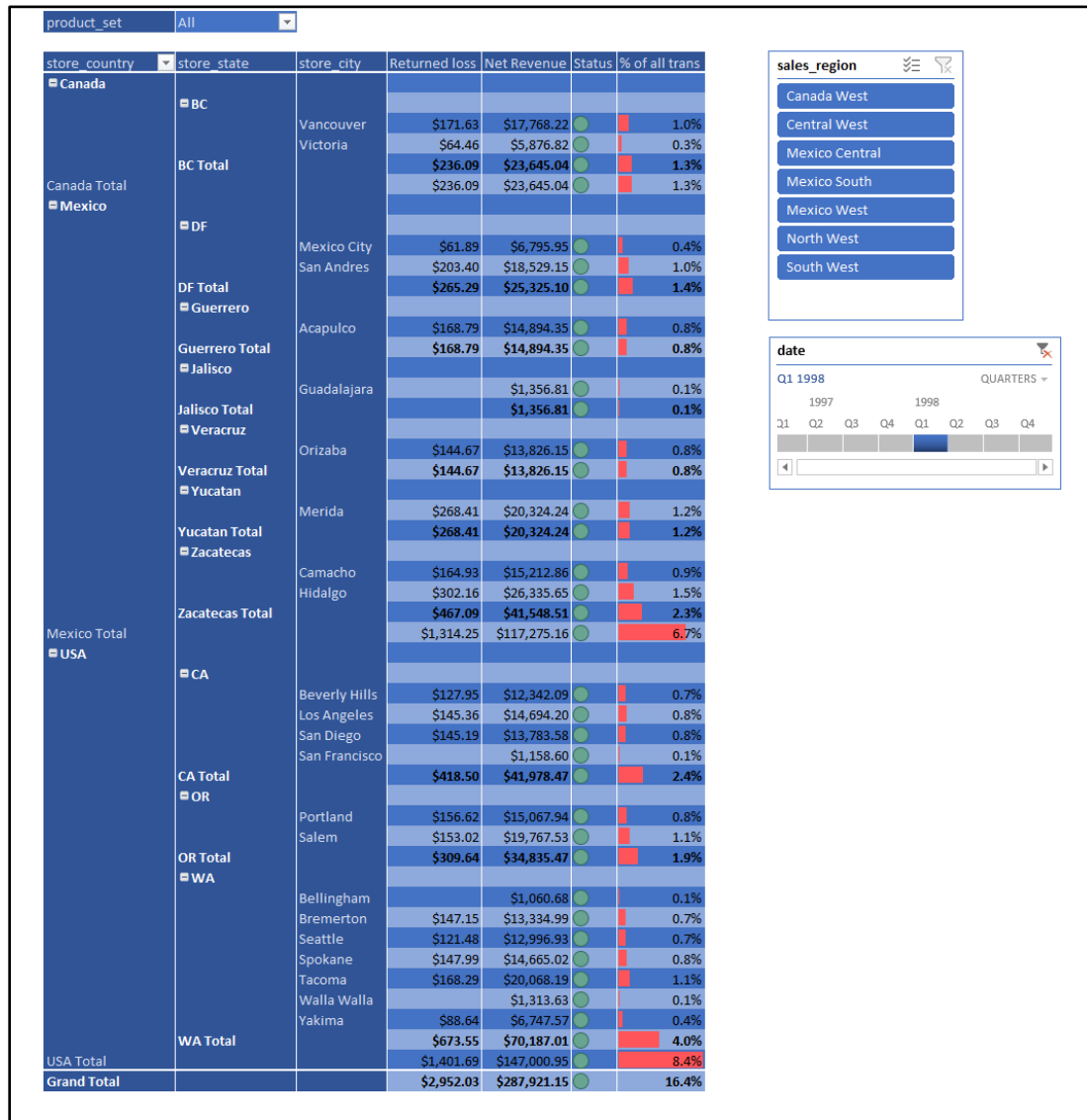
This example demonstrates the use of an Excel Pivot Table to analyze and present financial data with enhanced interactivity and clarity. The key features include:

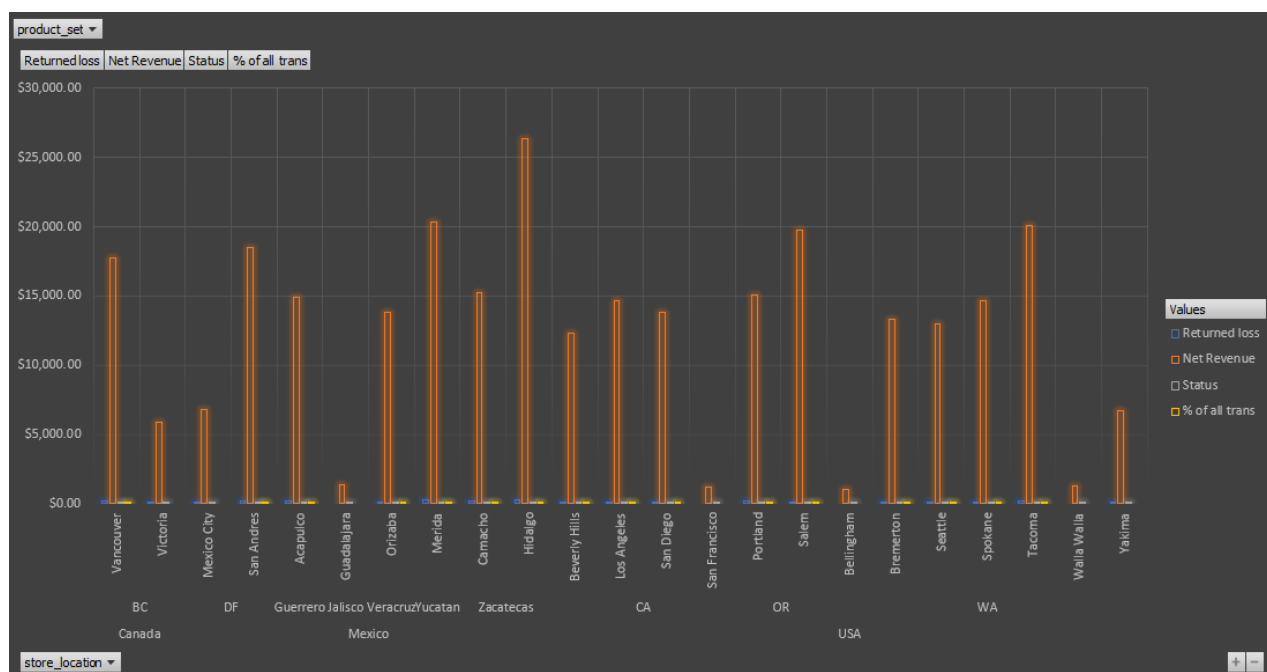
- **Metrics Analyzed:**
 - **Net Revenue:** Calculated as the difference between total transactions and returns.
 - **Percentage of Transactions:** Analyzed by applying various filters to understand transactional distribution.
- **Filters and Slicers:**
 - A **Slicer** is applied to segment data by sales region, enabling quick and intuitive filtering.
 - A **Timeline** filter is configured for Quarter-to-Date (QTD) analysis, streamlining the focus on relevant time periods.

- **Bar Chart:**

- An associated bar chart visually represents key metrics like net revenue and percentage of transactions. This provides a clear and comparative view of data across dimensions such as regions or time periods.

These tools collectively transform raw data into actionable insights, offering users the flexibility to explore and analyze results dynamically.





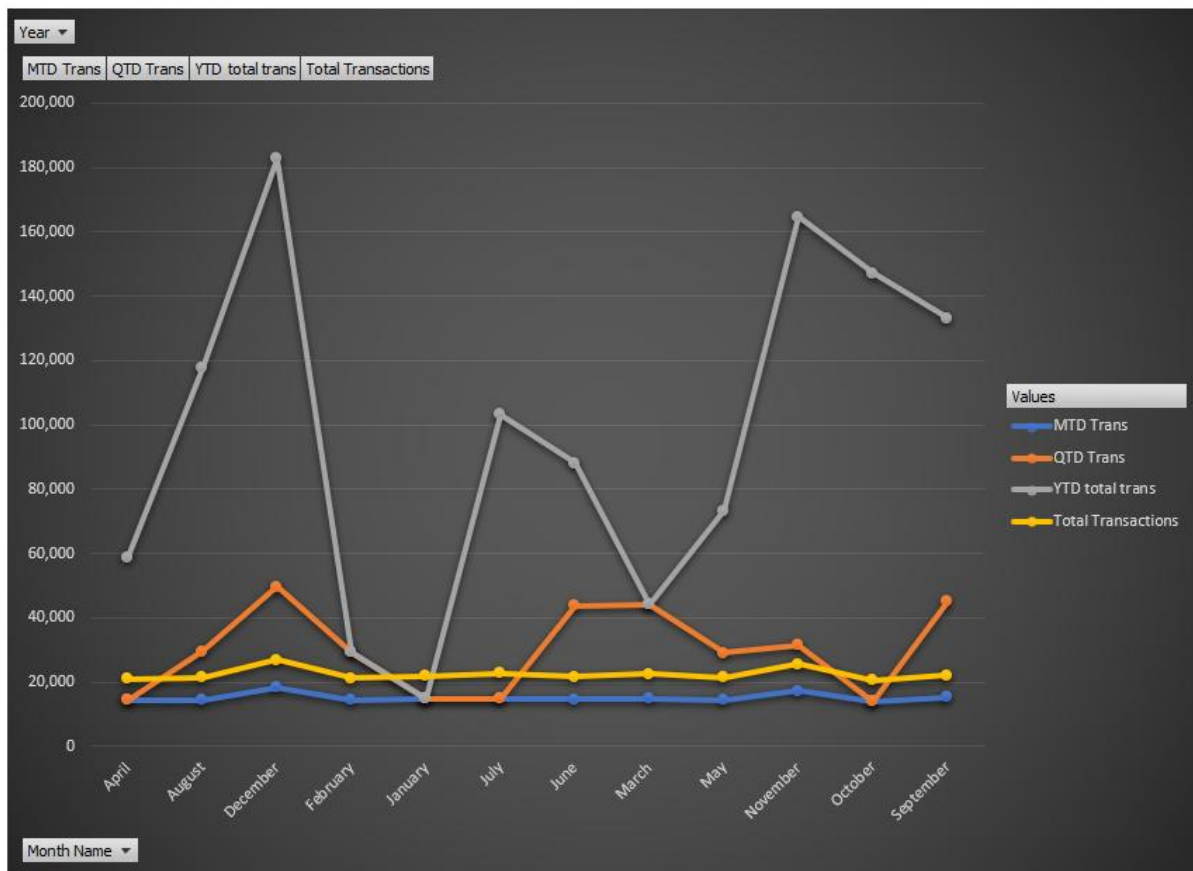
Excel Pivot Table Utilizing Time Intelligence with DAX Measures

This example highlights the use of an Excel Pivot Table to perform advanced time intelligence analysis, leveraging DAX (Data Analysis Expressions) measures.

Key Features:

- **Time Intelligence Metrics:**
 - **Month-to-Date (MTD):** Displays aggregated values for the current month up to the selected date.
 - **Quarter-to-Date (QTD):** Summarizes data for the current quarter up to the selected date.
 - **Year-to-Date (YTD):** Provides cumulative values for the current year up to the selected date.
 - **Total Transactions:** Calculates overall transaction totals for the selected period.
- **Interactive Filters:**
 - Slicers for Year and Quarter: Enable quick and intuitive filtering, allowing users to isolate specific time periods and analyze data trends seamlessly.
- **Pivot Chart:**
 - A dynamic visual representation of the data is generated alongside the Pivot Table, displaying key metrics.

Year All					date (Year)		Quarter	
Row Labels	MTD Trans	QTD Trans	YTD total trans	Total Transactions				
April	14,552	14,552	58,804	21,142	1997		1	
August	14,517	29,561	117,662	21,555	1998		2	
December	18,325	49,789	182,883	27,042	(blank)		3	
February	14,392	29,328	29,328	21,236			4	
January	14,936	14,936	14,936	21,970			(blank)	
July	15,044	15,044	103,145	22,796				
June	14,741	43,849	88,101	21,653				
March	14,924	44,252	44,252	22,634				
May	14,556	29,108	73,360	21,422				
November	17,339	31,464	164,558	25,571				
October	14,125	14,125	147,219	20,604				
September	15,432	44,993	133,094	22,095				
Grand Total	18,325	49,789	182,883	269,720				



Examples of some of the DAX measures implemented to create the time intelligence calculations:

Measure

Table name: Calendar_Lookup

Measure name: QTD Trans

Description:

Formula: Check formula

`=CALCULATE([Total Transactions], DATESQTD(Calendar_Lookup[date]))`

Formatting Options

Category:

- General
- Number
- Currency
- Date
- TRUE/FALSE

Format: Whole Number

☒ Use 1000 separator (,)

OK Cancel

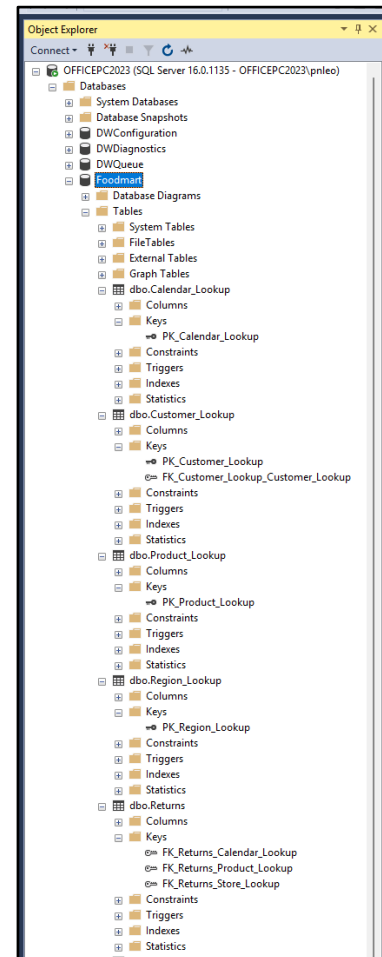
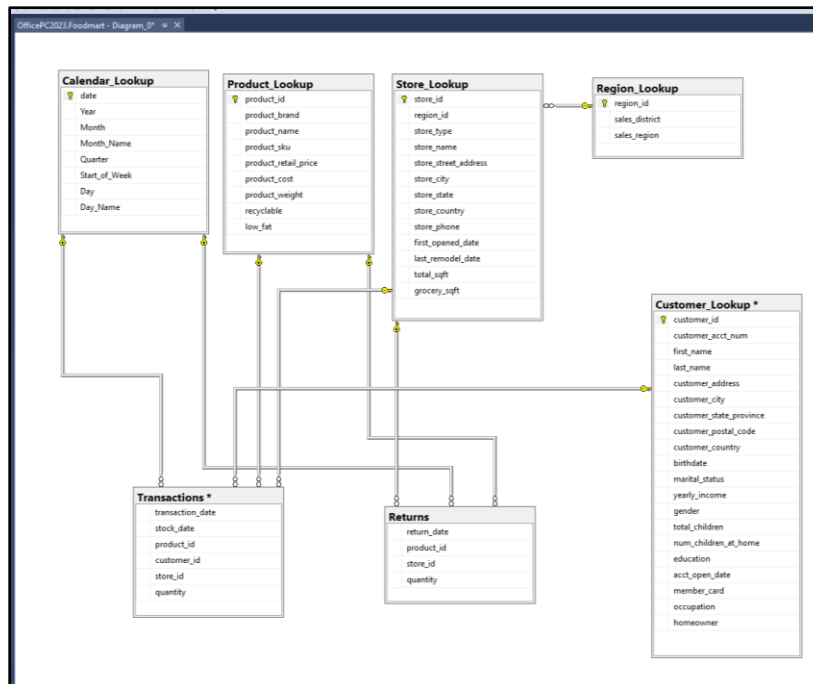
Measure	Formula
% of all trans	[Total Transactions]/[all transactions]
10 day average of tra...	CALCULATE([Total Transactions], DATESINPERI...
10 day rolling trans	CALCULATE([Total Transactions], DATESINPERI...
all transactions	CALCULATE([Total Transactions], ALL(transactions))
last month trans	CALCULATE([Total Transactions], DATEADD(Cale...
Month over Month Trans % ...	(([Total Transactions]-[last month trans])/[last month
MTD Trans	CALCULATE([Total Transactions], DATESMTD(Ce...
Net Revenue	[Total Revenue (Measure)]-[Returned loss]
product rank (by revenue)	RANKX(ALL(Product_Lookup), [Total Revenue (M...
QTD Trans	CALCULATE([Total Transactions], DATESQTD(Ce...
Quantity Returned	SUM(Returns[quantity])
Recyclable Products	COUNTA(Product_Lookup[recyclable])
Return Rate	[Quantity Returned]/[Total Quantity]
Returned loss	SUMX(Returns,>Returns[quantity]*RELATED(Produ...
threshold selection	max(price_threshold[price_threshold])
Total Quantity	SUM(transactions[quantity])
Total Revenue (Measure)	SUMX(transactions, [quantity] * RELATED(Product...
Total Transactions	COUNTROWS(transactions)
transaction under threshold	CALCULATE([Total Transactions], FILTER(Product...
Unique Products	DISTINCTCOUNT(Product_Lookup[product_id])

SQL Server database

Visualizing Table Relationships in SQL Server 2022 Express

The screenshot illustrates the table relationships established within SQL Server 2022 Express, managed via SQL Server Management Studio. It provides a dual view to offer comprehensive insights into the database's structure and relational integrity:

- **Object Explorer (right):** Displays the database structure, including the hierarchy of tables and their associated primary and foreign keys. This detailed view facilitates navigation and highlights key components of the database schema.
- **Database Diagram (below):** Offers a visual representation of the table relationships, complete with cardinality to describe the connections between tables. This diagram aids in understanding the flow of data and ensures accurate relational mapping within the database.



Together, these tools provide a clear and organized view of the database, supporting the creation and analysis of robust relationships within the dataset.

Here's an example of a **SQL SELECT** query using an **INNER JOIN** and a **filter**:

Sample SQL Query:

This SQL query retrieves specific data from four related tables (*Transactions*, *Product_Lookup*, *Store_Lookup* and *Customer_Lookup*), combining rows where there is a matching condition (*customer_id*, *product_id* and *store_id*) and then applying a date filter ('1998-01-03') to narrow down the results.

```

1 USE Foodmart
2 GO
3
4 SELECT
5     [dbo].[Transactions].[transaction_date]
6     ,[dbo].[Transactions].[quantity] as "Product QTY"
7     ,[dbo].[Product_Lookup].[product_name] as "Product"
8     ,[dbo].[Customer_Lookup].[first_name] as "Customer First Name"
9     ,[dbo].[Customer_Lookup].[last_name] as "Customer Last Name"
10    ,[dbo].[Customer_Lookup].[customer_acct_num]
11    ,[dbo].[Store_Lookup].[store_name]
12    ,[dbo].[Store_Lookup].[store_city]
13    ,[dbo].[Store_Lookup].[store_state]
14
15 FROM [dbo].[Transactions] Inner Join [dbo].[Customer_Lookup]
16 ON [dbo].[Transactions].[customer_id] = [dbo].[Customer_Lookup].[customer_id]
17
18 Join [dbo].[Store_Lookup]
19 ON [dbo].[Transactions].[store_id] = [dbo].[Store_Lookup].[store_id]
20
21 Join [dbo].[Product_Lookup]
22 ON [dbo].[Transactions].[product_id] = [dbo].[Product_Lookup].[product_id]
23
24 Where Year(transaction_date) = 1998 AND Month(transaction_date) = 1 AND Day(transaction_date) = 3
25 ORDER BY [dbo].[Store_Lookup].[store_city] DESC
26 GO
27

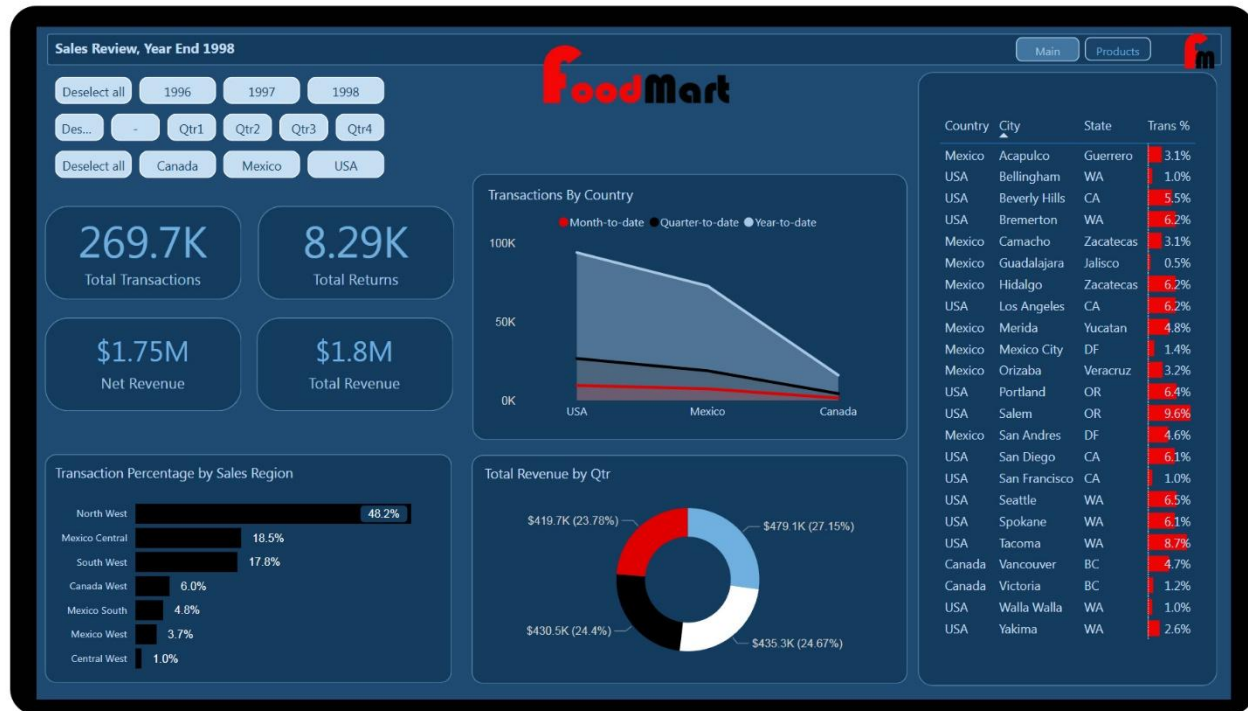
```

Excel Export of Results:

	A	B	C	D	E	F	G	H	I
1	transaction_date	Product QTY	Product	Customer First Name	Customer Last Name	customer_acct_num	store_name	store_city	store_state
2	1/3/1998	1	Hilltop Mint Mouthwash	Karen	Theisen	94908641576	Store 22	Walla Walla	WA
3	1/3/1998	3	Nationeel Fondue Mix	Harold	Gray	27933238700	Store 22	Walla Walla	WA
4	1/3/1998	2	High Top Mushrooms	Harold	Gray	27933238700	Store 22	Walla Walla	WA
5	1/3/1998	2	Denny Copper Pot Scrubber	Harold	Gray	27933238700	Store 22	Walla Walla	WA
6	1/3/1998	1	Portsmouth Merlot Wine	Harold	Gray	27933238700	Store 22	Walla Walla	WA
7	1/3/1998	2	Just Right Rice Soup	Harold	Gray	27933238700	Store 22	Walla Walla	WA
8	1/3/1998	2	Better Chicken Soup	Gerri	Young	35876799225	Store 22	Walla Walla	WA
9	1/3/1998	1	Discover Spaghetti	Gerri	Young	35876799225	Store 22	Walla Walla	WA
10	1/3/1998	1	Sunset Screw Driver	Gerri	Young	35876799225	Store 22	Walla Walla	WA
11	1/3/1998	2	Hermanos Macintosh Apples	Gerri	Young	35876799225	Store 22	Walla Walla	WA
12	1/3/1998	1	Tri-State Sweet Peas	Gerri	Young	35876799225	Store 22	Walla Walla	WA
13	1/3/1998	2	Even Better Cheese Spread	Gerri	Young	35876799225	Store 22	Walla Walla	WA
14	1/3/1998	2	Sunset Paper Cups	Mark	Manakoff	65586947200	Store 22	Walla Walla	WA
15	1/3/1998	2	Fort West Fudge Cookies	Mark	Manakoff	65586947200	Store 22	Walla Walla	WA
16	1/3/1998	2	Plato Hot Chocolate	Mark	Manakoff	65586947200	Store 22	Walla Walla	WA
17	1/3/1998	1	Pleasant Canned Tuna in Water	Mark	Manakoff	65586947200	Store 22	Walla Walla	WA
18	1/3/1998	1	Big Time Lime Popsicles	Mark	Manakoff	65586947200	Store 22	Walla Walla	WA
19	1/3/1998	2	Nationeel Buttered Popcorn	Kathy	Bee	75636937618	Store 22	Walla Walla	WA
20	1/3/1998	2	Landslide Creamy Peanut Butter	Kathy	Bee	75636937618	Store 22	Walla Walla	WA
21	1/3/1998	2	Better Canned Peas	Kathy	Bee	75636937618	Store 22	Walla Walla	WA

Power BI Desktop Reports

Main Dashboard:

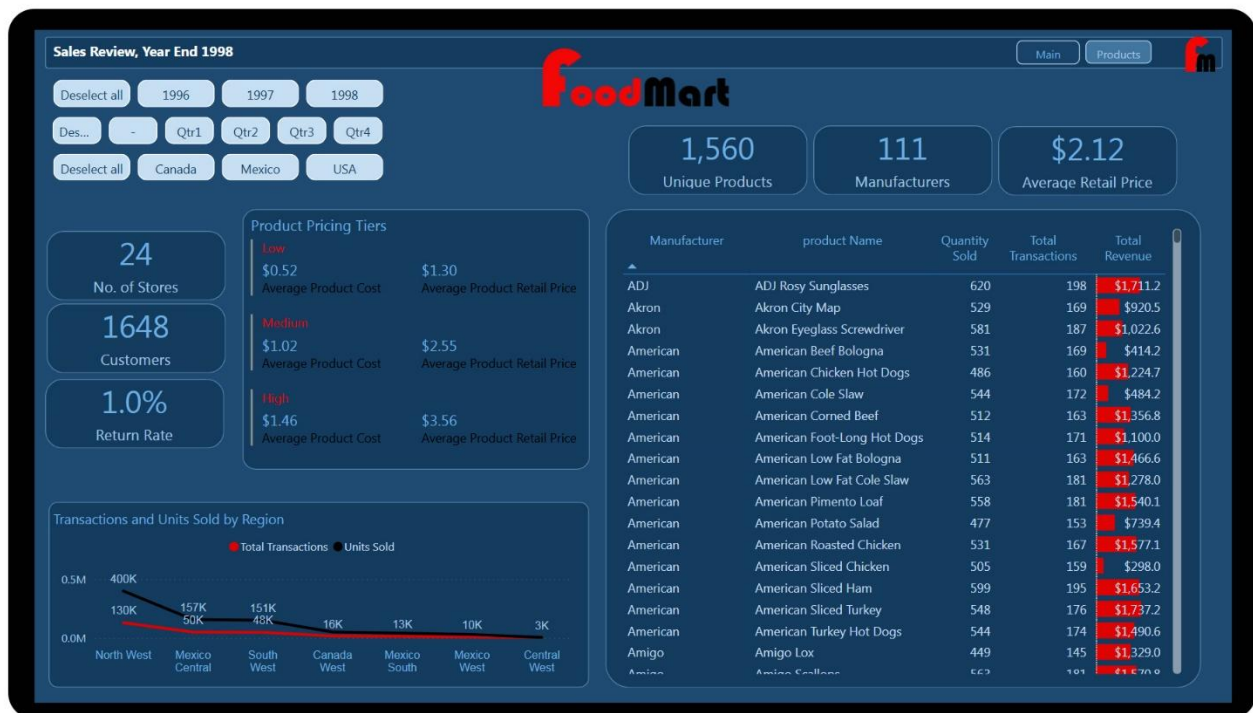


The Power BI dashboard presents key performance metrics and visualizations for Foodmart's sales data. Here's a breakdown of its elements:

- **Filters:**
 - Year filter: 1996, 1997, 1998
 - Quarter filter: Q1 - Q4
 - Country filter: Canada, Mexico, and the USA
- **Metrics Displayed:**
 - **Total Transactions:** 269.7K
 - **Total Returns:** 5.65K
 - **Net Revenue:** \$1.19M
 - **Total Revenue:** \$1.2M
- **Visualizations:**
 - **Line Graph:** "Transactions by Country" tracks metrics like Month-to-Date (MTD), Quarter-to-Date (QTD), and Year-to-Date (YTD) for the USA, Mexico, and Canada.

- **Bar Chart:** "Transaction Percentage by Sales Region" illustrates the distribution of transactions across regions such as North West, Mexico Central, South West, Canada West, and others.
- **Pie Chart:** "Total Revenue by Quarter" shows revenue distribution across quarters (Q1: \$287.0K, Q2: \$326.4K, Q3: \$290.9K, Q4: \$326.4K).
- **Table:**
 - A table lists transaction percentages broken down by city, state, and country, allowing for granular insights.

Product Dashboard:



- **Filters:**
 - Year filter: 1996, 1997, 1998
 - Quarter filter: Q1 - Q4
 - Country filter: Canada, Mexico, and the USA
- **Metrics Displayed:**
 - **Stores:** 24 total
 - **Customers:** 1,648 total
 - **Return Rate:** 1.0%
 - **Products:** 1,560 unique products
 - **Manufacturers:** 111 total
 - **Average Retail Price:** \$2.12
- **Product Pricing Tiers:**

- **Low Tier:** Average cost \$0.52, retail price \$1.30
- **Medium Tier:** Average cost \$1.02, retail price \$2.55
- **High Tier:** Average cost \$1.46, retail price \$3.56
- **Detailed Sales Data:** Includes columns for Manufacturer, Product Name, Quantity Sold, Transactions, and Total Revenue.
- **Regional Performance:**
 - The dashboard visualizes sales data by region with metrics like Total Transactions and Units Sold. For example:
 - **North West:** 130K transactions, 400K units sold.
 - **Mexico Central:** 157K transactions, 50K units sold.
 - Other regions like South West, Canada West, Mexico South, and others are also included with specific transaction and unit details.

The dashboards provide a comprehensive view of sales performance, enabling the analysis of trends and actionable insights to improve decision-making.